



BACHELOR THESIS

RAW: a GitLab Integration within Moodle

Department

TIC

Program

Computer Science and Communications Systems

Specialization

Software Engineering

Student

Jouve Léonard

Date

17 August 2026

Proposed and supervised by

Delgado Pamela · HEIG-VD

Context

LMS and VCS platforms are both essential in computer science education, but operate independently, leading to fragmented workflows.

Teachers

- Create repositories manually
- Configure repository permissions by hand
- Manage multiple platforms

Students

- Inconsistent workflows between courses
- Switch between LMS and VCS platforms
- Grades, deadlines, repositories in different places

Platforms Involved

Learning Management System (LMS)

- Centralizes course organization and documents
- Manages assignments, submissions, and grading
- Structured teaching environment

Version Control System (VCS)

- Manages source code and development workflows
- Enables collaboration through Git repositories
- Provides project management / DevOps tools



Moodle

Modular, open-source, institutionally established



GitLab

Self-hosting, data sovereignty, API and webhooks

Specification

Objectives

- Assess existing solutions and APIs
- Design a modular architecture
- Implement, deploy, and validate

Expected Result

- Extensible Moodle-GitLab integration
- Automated, centralized workflows
- Simple, open-source solution

Deliverables

- Report and planning
- Presentation and poster
- Plugins, documentation, tests, and architecture

Design Principles

Design a solution around 3 core principles

Automation

- Automate GitLab resource management
- Favor flexibility over rigid workflows

LMS-native

- Provide a unified workflow
- Central access point over fragmented platforms

Extensible Design

- Open-source allowing contributions
- Modular architecture for futur extensions

Proprietary Solutions Risks

On Friday August 28th GitHub Classroom will fully transition to partner solutions, [...] and the GitHub Classroom website will be decommissioned. [...] We know that change is not easy and many of you have been relying on GitHub Classroom for years [...].

The GitHub Education Team

- Risk of service discontinuation
- Potential pricing changes
- Limited adaptability to specific needs
- Vendor lock-in

State of the Art

	Automation	Open source	LMS integration	VCS independent
GitHub Classroom	✓	●	✗	✗
Codio	✓	✗	●	✗
Classroom 50	✓	✓	✗	✗
RepoBee	●	✓	✗	✓
Classmoji	✓	✓	✓	●
RAW	✓	✓	✓	●



Repository Assignment Workflow (RAW)

A Moodle-native, extensible, and open-source solution.

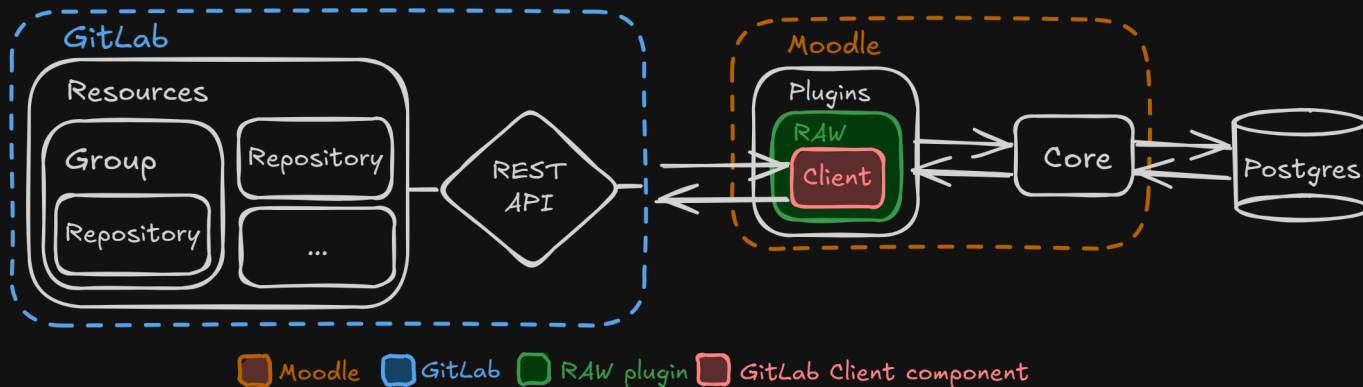
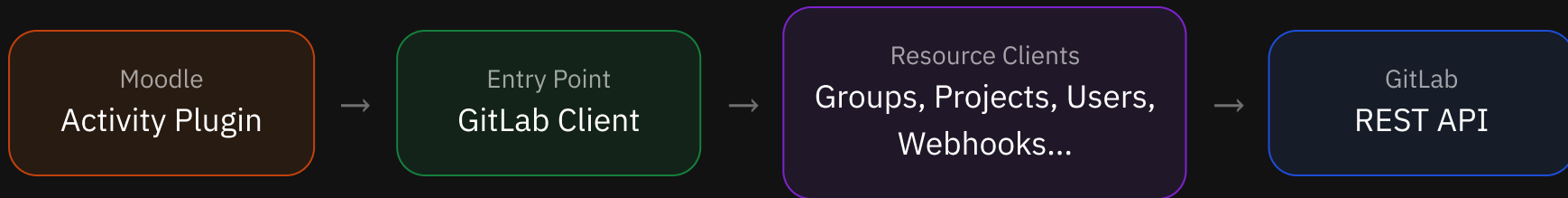


Figure 1. RAW architecture diagram

- Follows the 3 design principles defined previously
- Built using two Moodle plugins: **Custom Field, Activity Module**

GitLab Client

Abstraction layer between Moodle and GitLab REST API

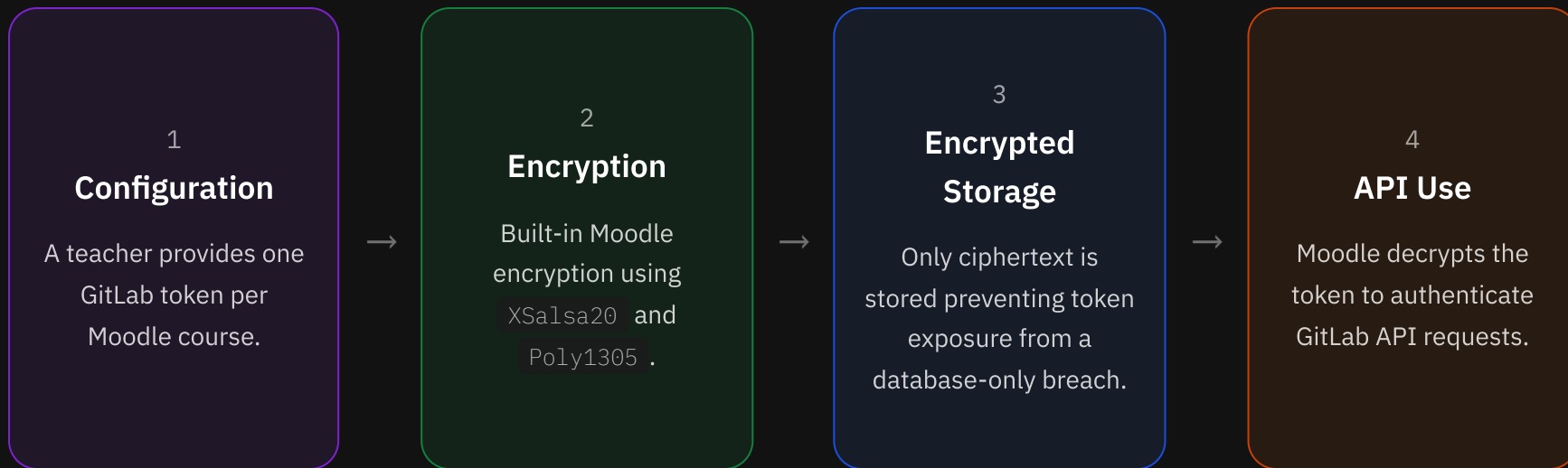


```
$client = new GitLabClient($token);  
$client->group()->create(...);  
$client->project()->fork(...);
```

- Provides high-level operations instead of raw HTTP requests
- Inspired by the GitLab PHP API Client library
- Resource clients mirror GitLab's API structure

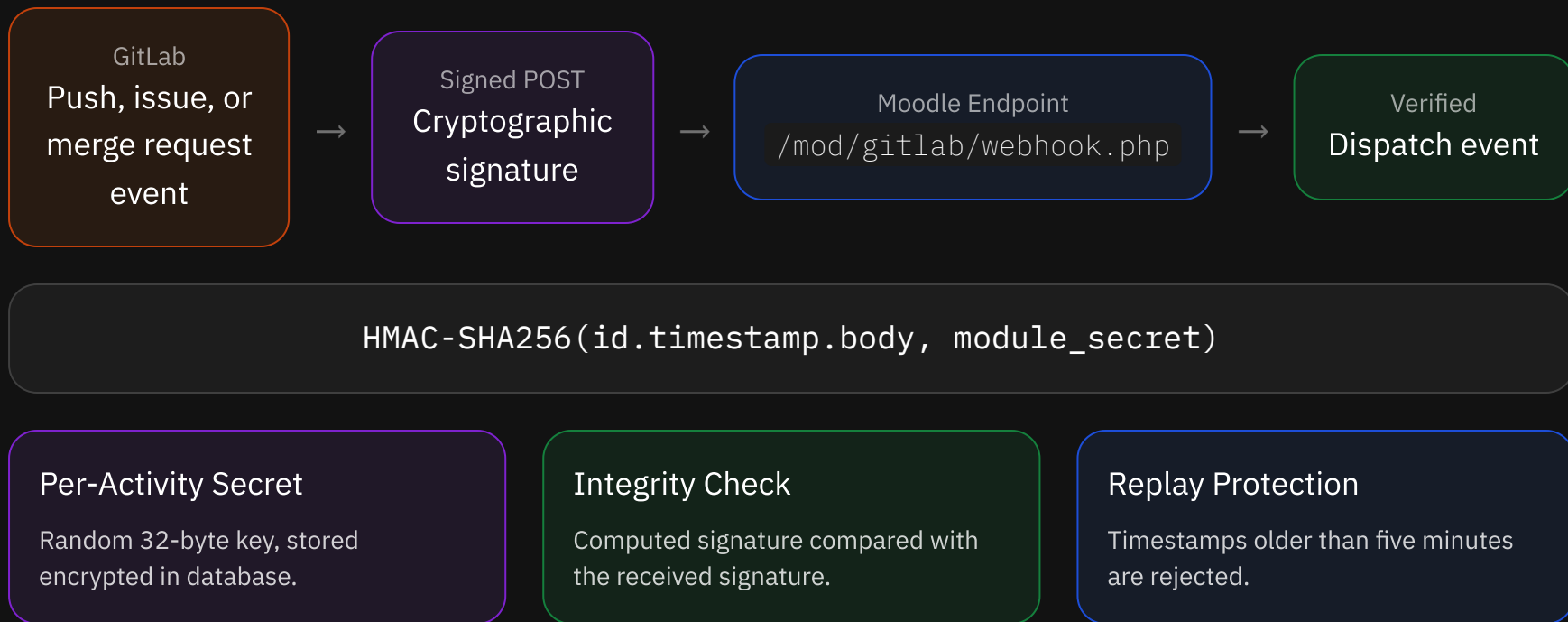
GitLab Tokens

Secure persistence of sensitive values with encryption



Webhooks

GitLab-to-Moodle synchronization through signed events



Only verified events can update templates, repositories, or submission status.



Demonstration

Features

Assignment Management

- RAW activity configuration
 - Due dates
 - Group capacity
 - Reviewers list
- Moodle calendar events and notifications
- Release solution access

User Workflows

- Teacher dashboard
 - Manage groups and members
 - Access template repository
 - Retrieve submissions
- Student interface
 - Create or join groups
 - Access repository information

Features

GitLab Integration

- Automated repository provisioning
- Repository access and permission management
- Webhook synchronization
- Protected files verification
- Custom GitLab host support

Template repository

- Instructions issue
- Solution branch
- Group submissions merge request

Group repository

- Fork of the template repository
- Instructions issue
- Submission merge request

Limitations and Future Work

Limitations

- Limited by GitLab's permission model
- Administrator access for plugin installation
- Dependence on a GitLab access token
- Manual Moodle–GitLab identity mapping

Future Work

- Improve error handling and user feedback
- Process long-running operations asynchronously
- Support additional Version Control Systems (VCS)
- Validate with representative users

Contributions

- Introduces a flexible, automated workflow
- Provides an extensible, open-source, Moodle-native integration
- Reduces manual coordination between LMS and VCS platforms
- Improves consistency and efficiency for instructors and students

